The Biosecurity Economics section within ABARES provides economic modelling research, cost-benefit analysis and related advice on biosecurity issues to the broader department and external clients. The section undertakes a range of multi-disciplinary research projects that assist ABARES, the department and external clients achieve strategic objectives as they relate to biosecurity.

Key capabilities of the section include economic analysis of the impacts of pest and disease incursions at the national, regional, industry and farm scales, and cost-benefit analysis of different response options to assist with emergency response planning and broader biosecurity decision making.

Research reports

This report outlines the process of climate matching using the Climatch V2.0 and the results of the potential distribution modelling for the Polyphagous shot-hole borer in Australia.

This technical report examines how changing regulations on the use of antimicrobial veterinary medicinal products and medicated feed could affect Australia's food animal producers

This report provides estimates on the cost of 2 plausible eradication scenarios for an African swine fever (ASF) incursion into Australia and the cost of endemic ASF.

This report provides an improved understanding of the impact of a Xylella fastidiosa incursion on the Australian economy and the environment.

This report has been prepared to provide evidence to assess the benefits from maintaining freedom from Equine Influenza and to help assessments over the costs and benefits of interventions should a new outbreak occur.

Starting in 2014–15, the Australian Government committed $8.96 million over six years to 149 grants to Rural Research and Development Corporations to invest in improving farmers' access to minor use chemicals for pest control. These grants were used to conduct field trials to generate the safety and efficacy data required for chemical registration in Australia.

This report estimates the benefits to industry of 15 case study grants. Overall, the program was found to generate positive returns to the Australian Government's investment and stakeholder contributions, with an average $117 returned over 20 years from each $1 of money invested.

Wheat stem rust is present in many wheat growing areas throughout the world, and around two-thirds of global wheat growing areas are climatically suitable for the disease.

Ug99, so-named because it was found in Uganda in 1999, is a particularly virulent strain of wheat stem rust that has overcome 17 out of 34 stem rust resistance genes found in wheat. Around 30 per cent of current wheat varieties show moderate to high susceptibility to the Ug99 strain.

This report estimates the economic impact of a wheat stem rust strain Ug99 outbreak on the Australian wheat industry.

Recent outbreaks of Xylella fastidiosa in Europe prompted Australia's biosecurity authorities to introduce emergency measures in late 2015 to reduce the risk of an incursion.

As part of the heightened preparedness, the department's Plant Biosecurity Division requested ABARES to assess the economic impacts of a potential X. fastidiosa outbreak on the wine grape and wine-making industries in Australia.

This ABARES report presents estimates of the economic impact of a hypothetical outbreak of scrapie in Australia's sheep and goat population under three disease spread scenarios:

- eradicable epidemic—an outbreak occurs and is successfully eradicated
- managed spread—eradication is unsuccessful and spread is slowed by control measures
- uncontrolled spread—without control measures in place, the disease spreads uncontrolled.

In this report, ABARES estimates the value of Australia's biosecurity system at the 'farm gate'. The report considers the effect on farm profits of an outbreak of six potentially significant biosecurity threats to Australian agriculture: foot-and-mouth disease (FMD), Mexican feather grass, citrus greening, avian influenza, Karnal bunt and red imported fire ants.

In this report, ABARES compares eradication and containment costs to identify the most cost-effective way to respond to a hypothetical incursion of black-striped mussel (Mytilopsis sallei) in Australia. This invasive mussel species is increasingly widespread in ports throughout Asia, from the west coast of India to the north coast of Japan, and many of the infested ports involve seaborne trade with Australia.

In this report, ABARES compares the cost-effectiveness of 14 control scenarios for red imported fire ants (RIFA) (Figure 1), an invasive ant species that is currently contained within South East Queensland but has the potential to spread throughout Australia if left unchecked.

The control scenarios considered include two different budgets for surveillance and treatment ($10.5 million and $21 million a year), the share of the budget allocated to remote sensing activities, and assumptions of high and low sensitivity of new remote sensing technology introduced to improve surveillance and nest detection.

In this report, ABARES assesses the economic costs and social impacts of a hypothetical outbreak of foot-and-mouth disease (FMD) in Australia. FMD is a highly contagious viral infection of cloven hoofed mammals, including cattle, sheep, goats, pigs, deer and camels. Overseas experience shows the disease has serious economic and social implications, particularly for countries producing and exporting livestock and livestock products.

In this report, ABARES presents a benefit-cost analysis framework to assess the economic feasibility of different biosecurity control measures in the event of a hypothetical incursion of Varroa in Australia. Factors considered in the modelling include the spread of Varroa from each of the ports in Sydney, Melbourne and Cairns, whether the spread is unhindered or contained, and the development of a managed pollination industry that increases supply of pollination services in response to increasing demand.

Source: https://www.agriculture.gov.au/abares/research-topics/biosecurity/biosecurity-economics